



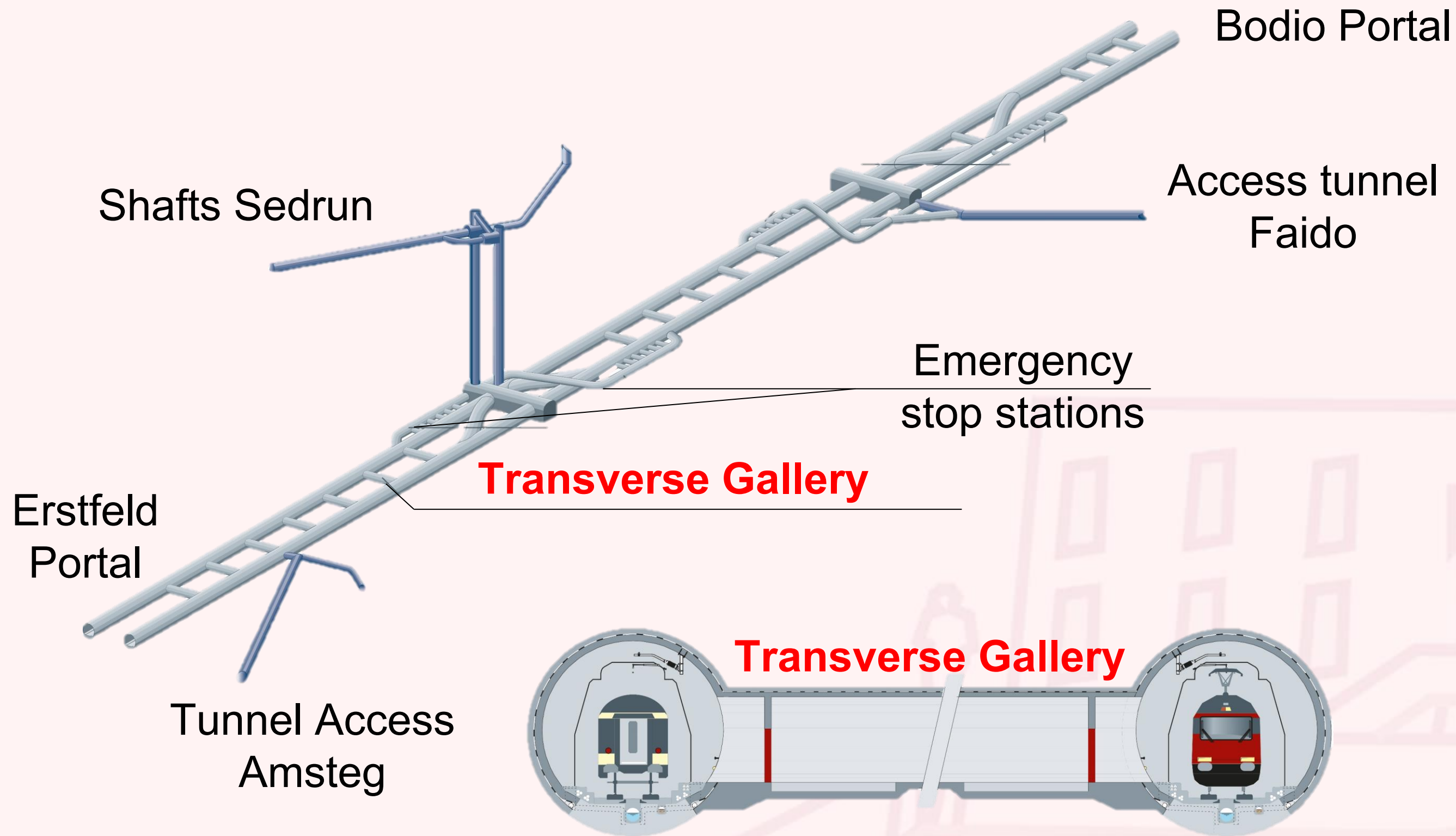
Pressure anisotropy on twin tunnel linings induced by a transverse gallery construction using time-dependent constitutive models

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Research problem



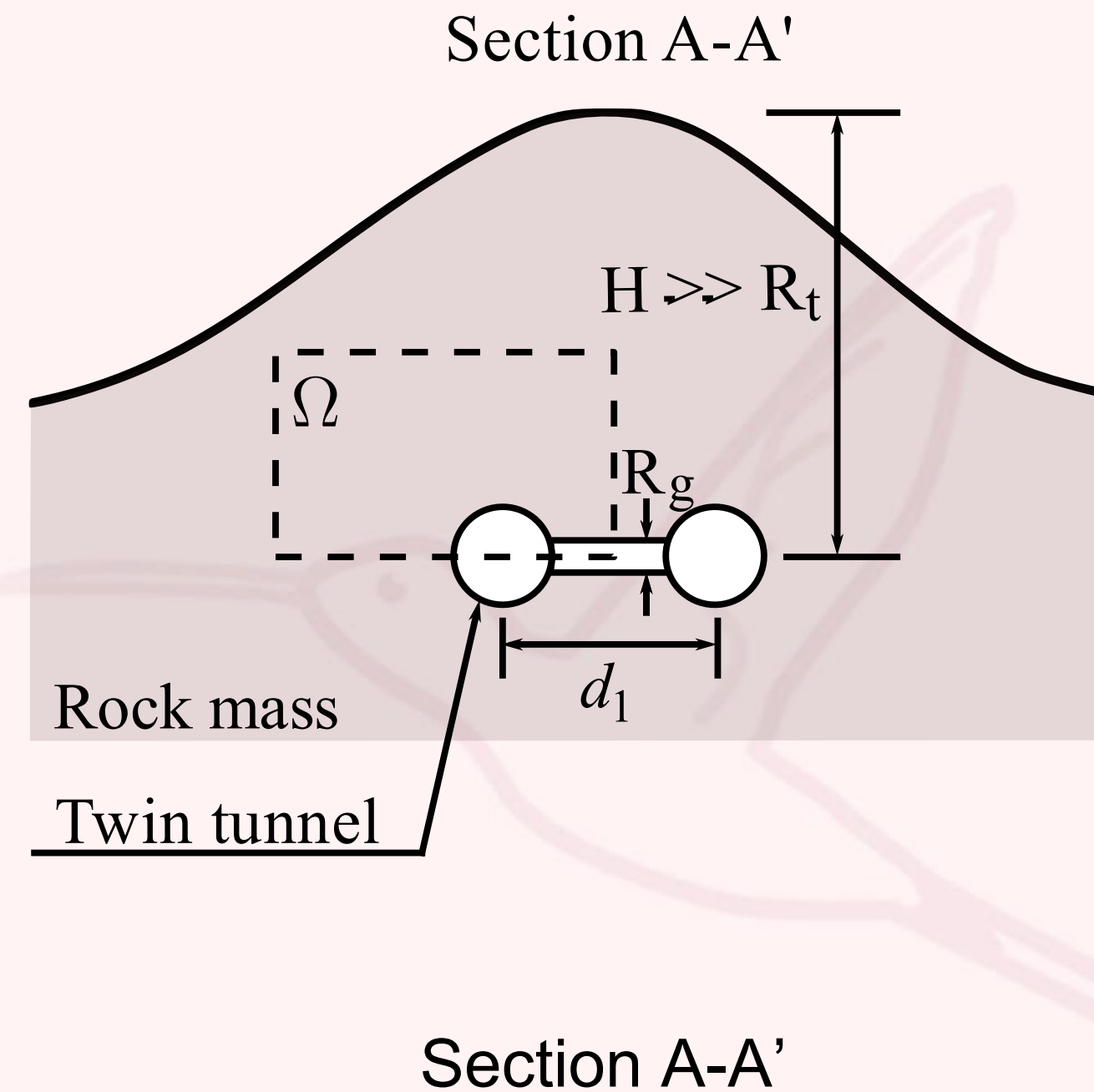
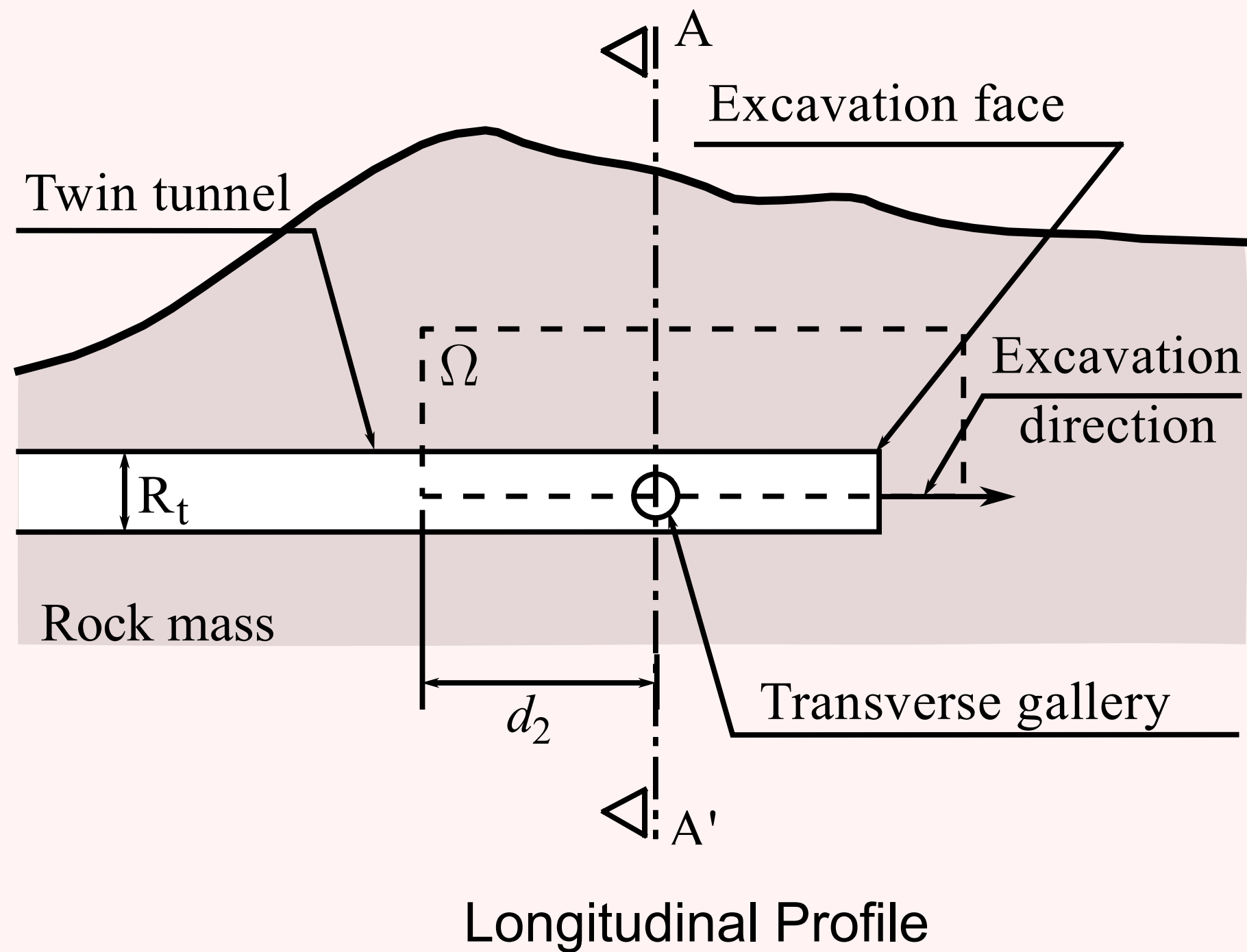
Objective

Present a study of the differences in the anisotropic redistribution of lining pressures due to the presence of the transverse galleries in twin tunnels comparing elastic and viscoelastic linings in elastoplastic-viscoplastic rockmass.

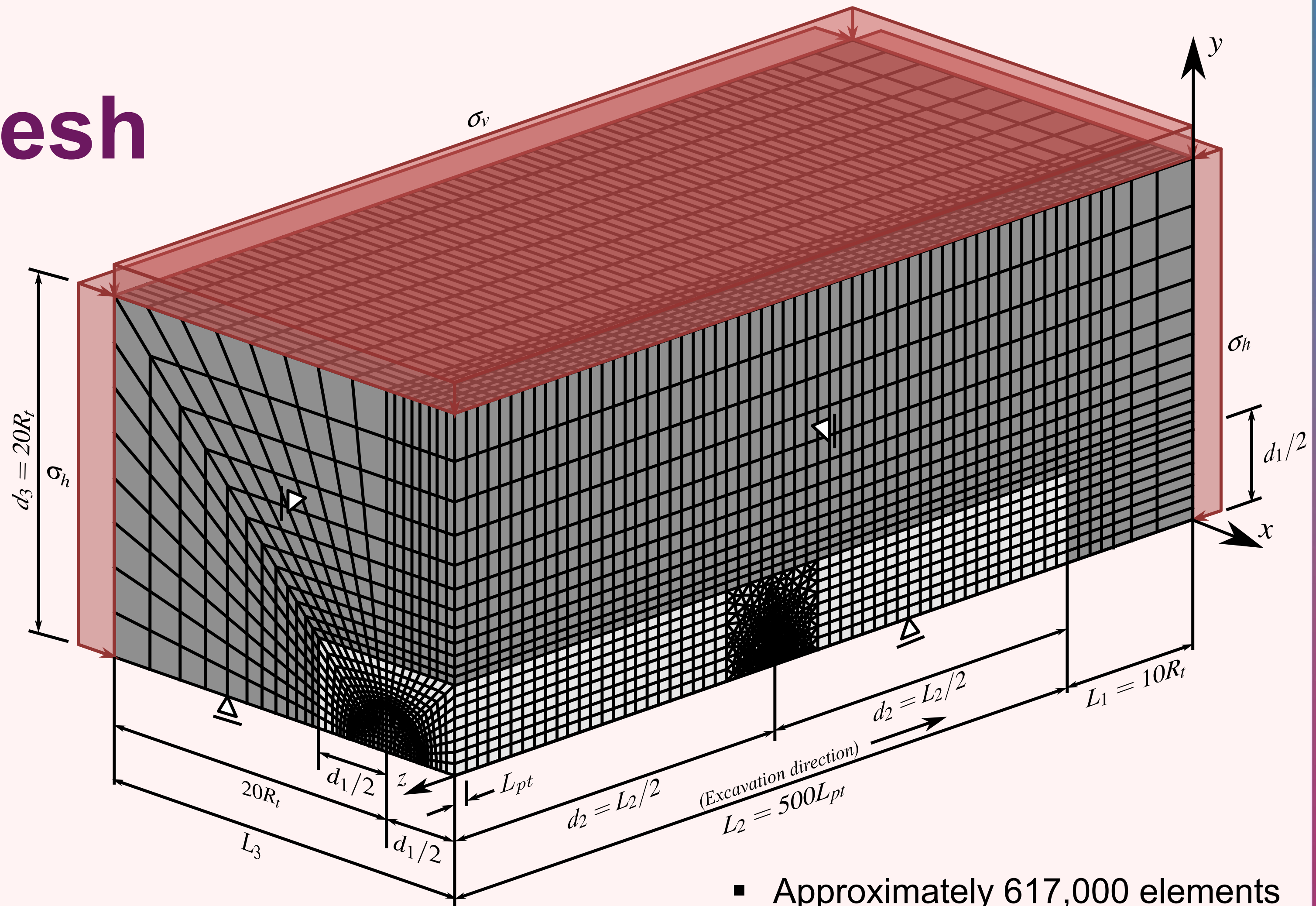


F.E Model

Schematic representation of the geometry problem:

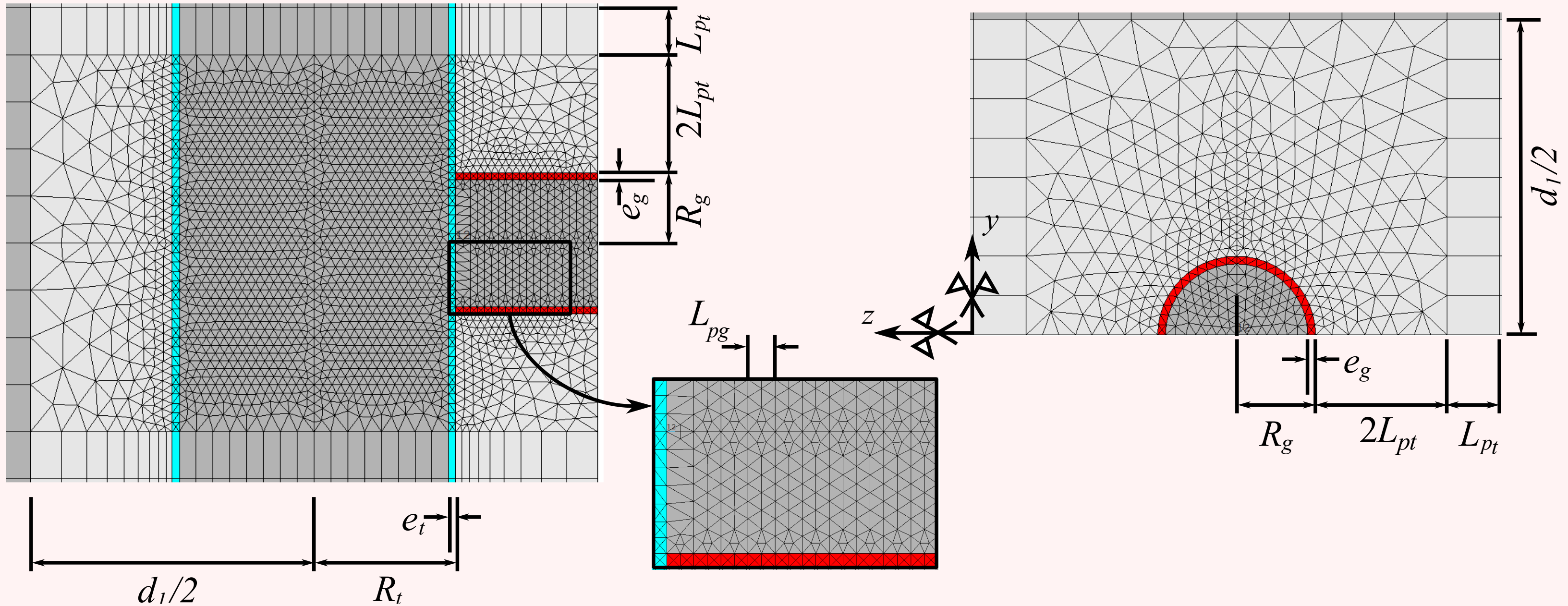


Mesh



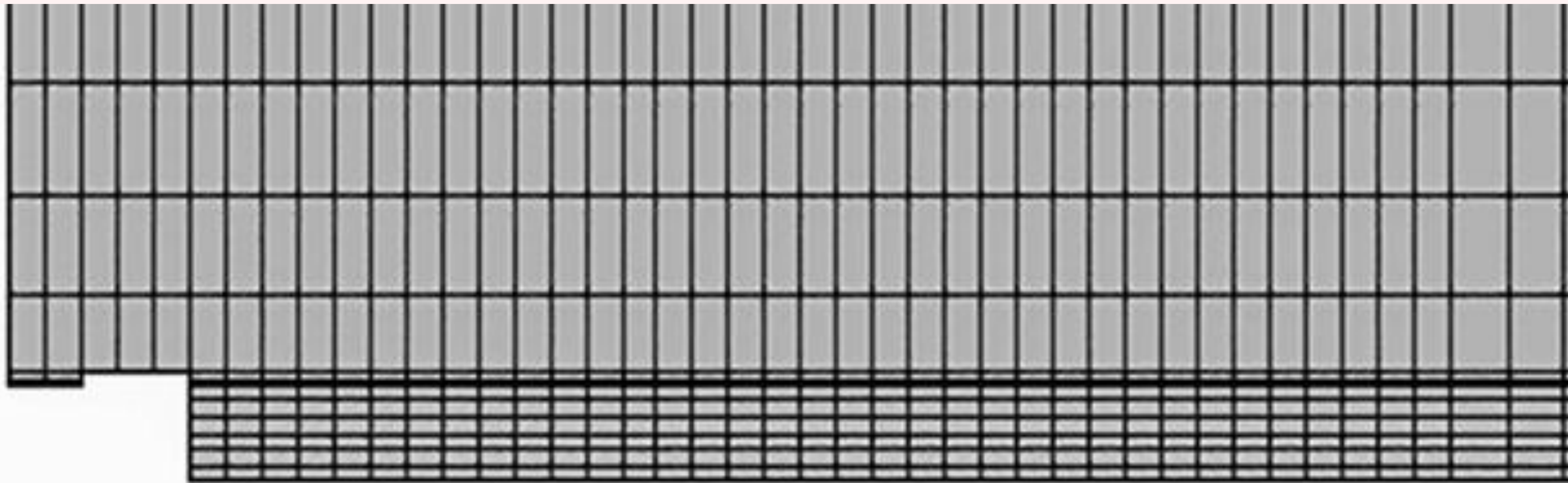
- Approximately 617,000 elements

Mesh



- 8-node trilinear hexahedral;
- 4-node linear tetrahedral;

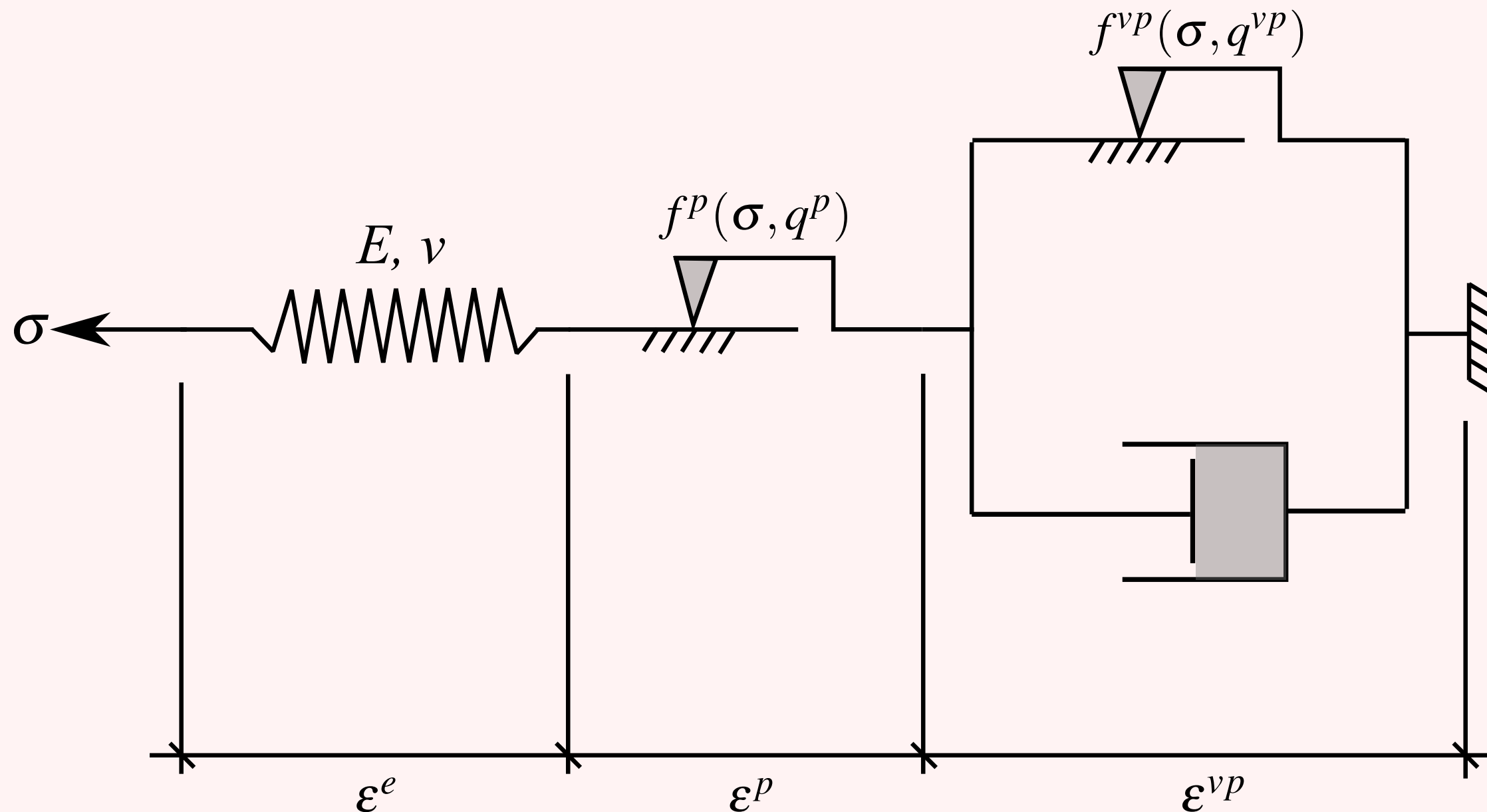
Activation and deactivation method



Materials constitutive models

Rockmass:

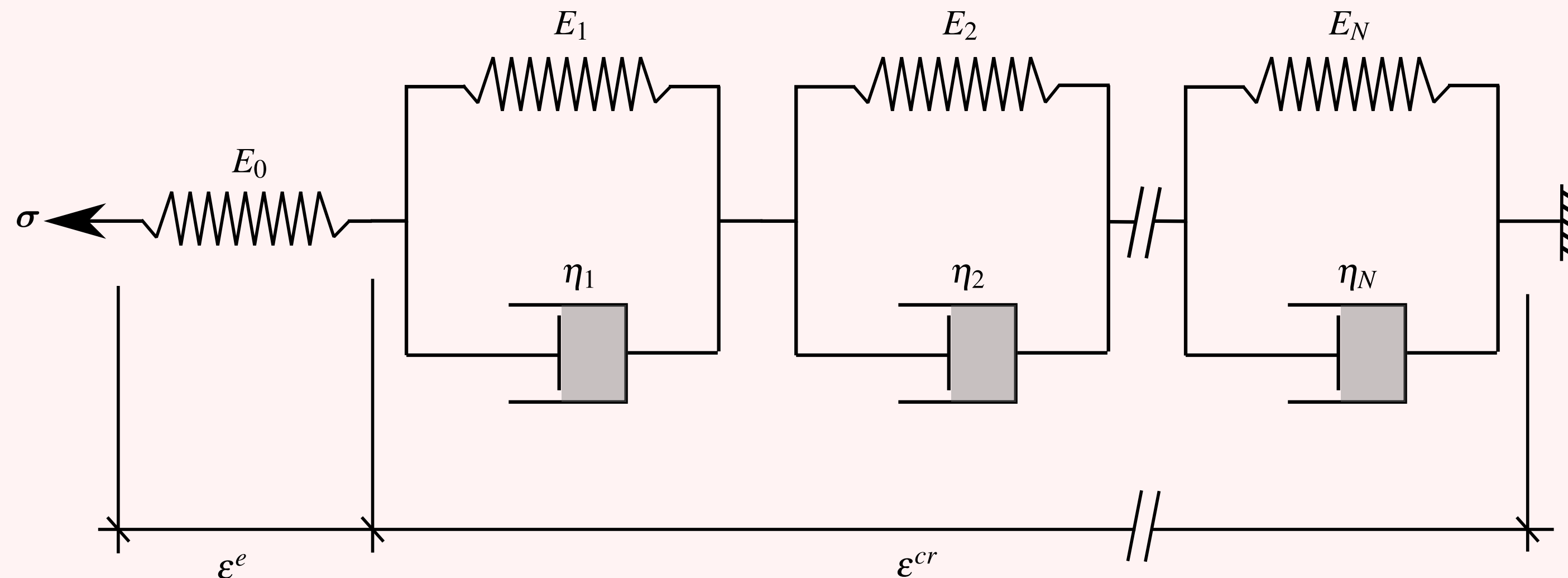
- Coupled elastoplastic-viscoplastic formulation by Rousset (1989) and Quevedo (2021);
- Drucker-Prager inscribed in Mohr-Coulomb.



Materials constitutive models

Lining:

- Elastic and Viscoelastic linings (Creep and Shrinkage) – Quevedo (2017)



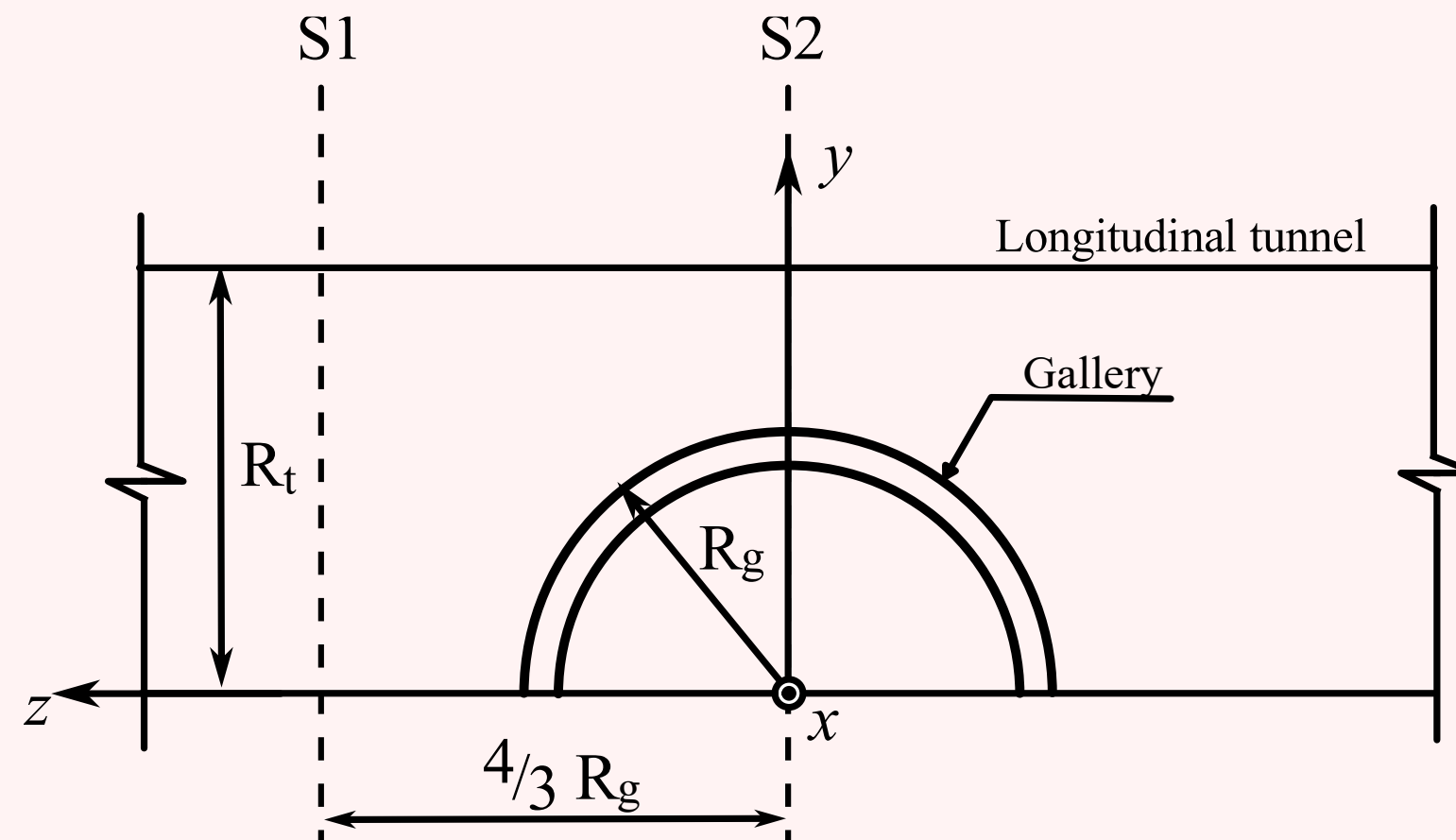
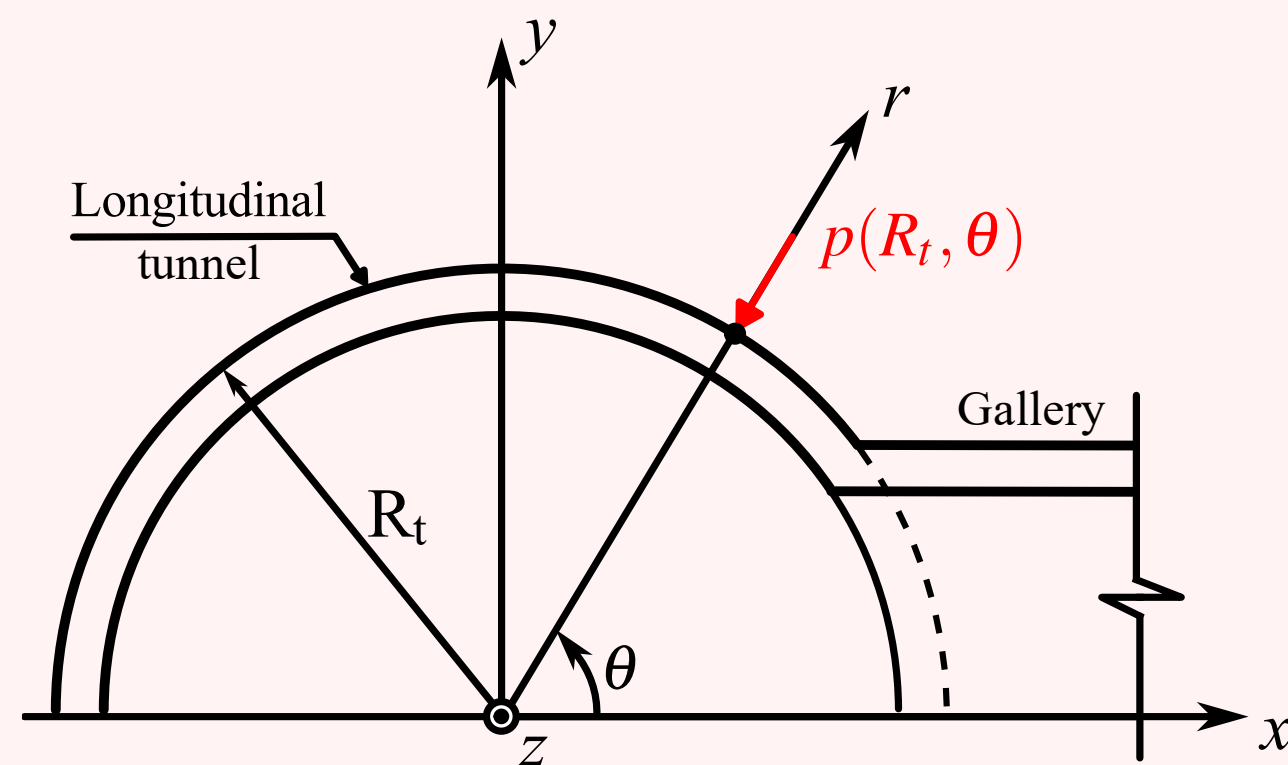
Schematic representation of Generalized Kelvin model for the concrete Bazant & Prasannan (1989) with parameters calibrated by CEB-FIP (1990) formulation

Three-dimensional finite element simulations

Numerical results of lining pressures are presented at the **end of construction** and **after 3,000 days**, representing long-term viscous stabilization.

The anisotropies of pressures are represented by:

$$\chi(\theta) = \frac{p(\theta)}{p(\theta = 90^\circ)}$$



Three-dimensional finite element simulations

Rockmass model data:

- $E_m = 1,400 \text{ MPa}$
- $\nu_m = 0,4$
- $c^p = 4 \text{ MPa}$
- $\phi^p = 0^\circ \text{ and } 5^\circ$
- $c^{vp} = 2 \text{ MPa}$
- $\phi^{vp} = 0^\circ$
- $\eta = 40,000 \text{ MPa} \cdot \text{day}$
- $\sigma_v = \sigma_h = 9 \text{ MPa}$

Lining model data:

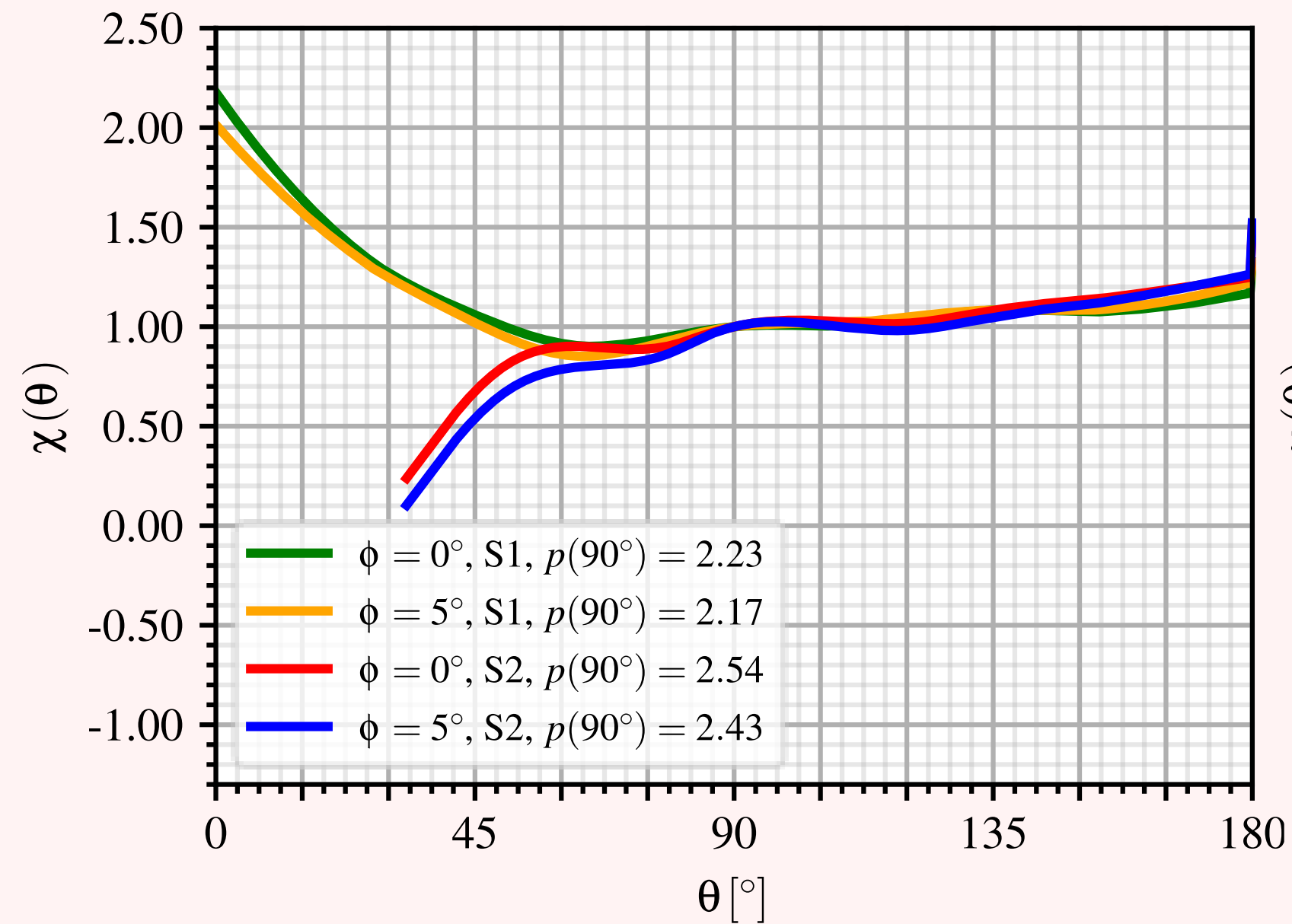
- $E_{c_{28}} = 25,146 \text{ MPa}$
- $\nu_c = 0,2$

Tunnel data:

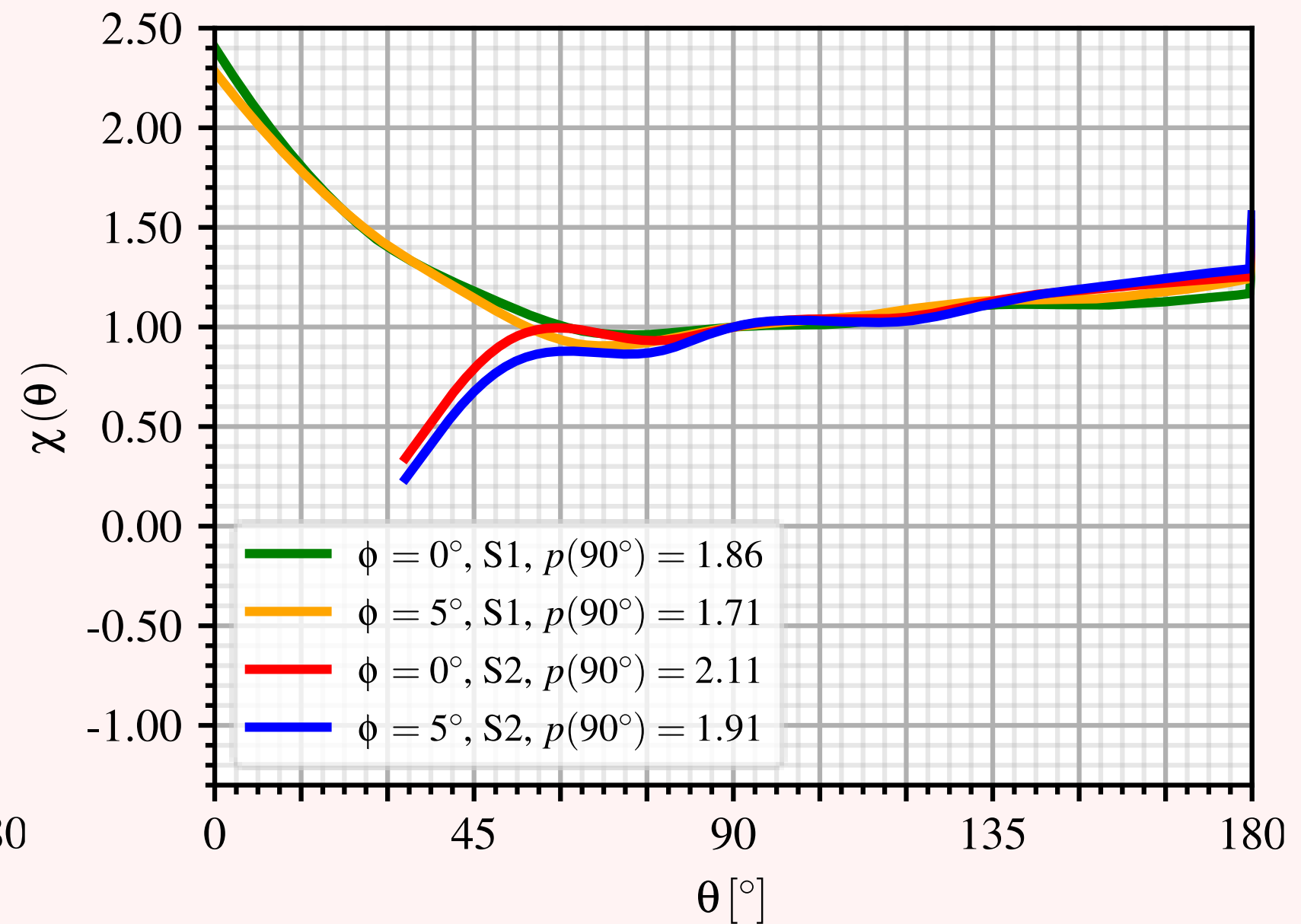
- $R_t/R_g = 0.5$
- $d_1 = 4R_t$
- $e_g = e_t = 0.05R_t$
- $d_0 = 2R_t/3$
- $L_p = R_t/3$
- $V_p = 12.5 \text{ m/s}$

Three-dimensional finite element simulations

Anisotropy of pressures in longitudinal tunnel lining at the **end of construction**:



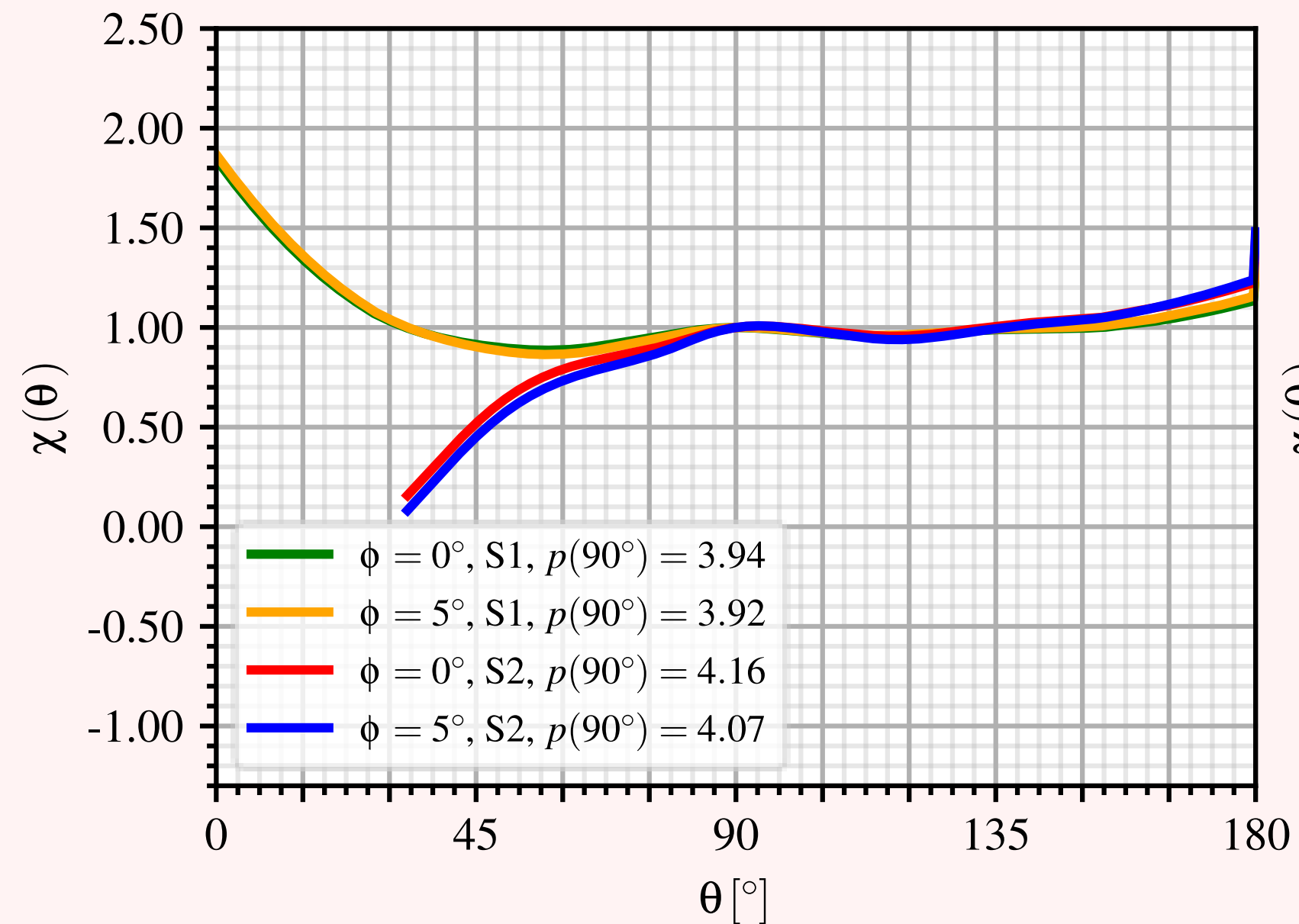
(a) Elastic lining



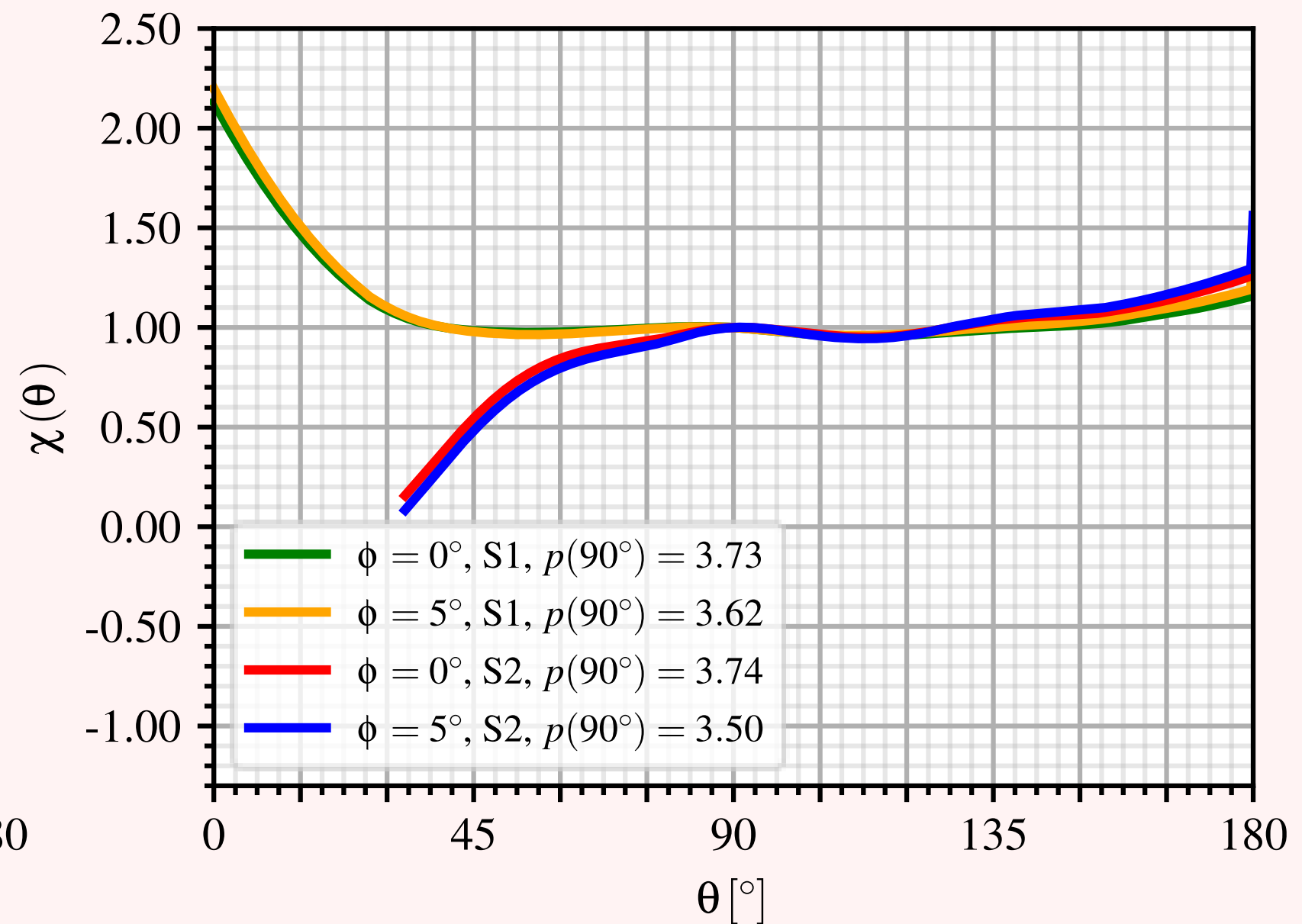
(b) Viscoelastic lining

Three-dimensional finite element simulations

Anisotropy of pressures in longitudinal tunnel lining at the **long-term**:



(c) Elastic lining

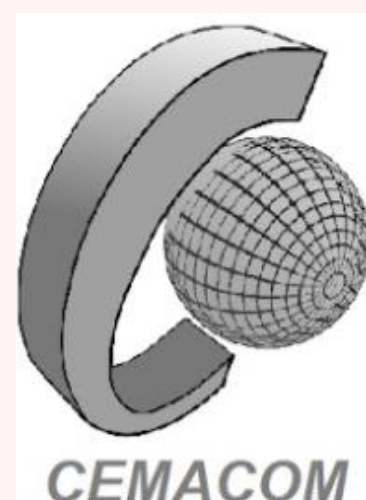


(d) Viscoelastic lining

Conclusions

- The transverse gallery causes a significant increase in lining pressures in the lateral tunnel wall near the parallel tunnel, particularly in section S1;
- Stress concentrations form around the gallery entrance, modifying the pressure distribution along the lining;
- Over the long term, pressures increase, but the influence of the friction angle becomes less pronounced;
- The viscoelastic lining shows greater anisotropy relative to the crown at $\phi = 0^\circ$ when compared to elastic concrete;
- These findings emphasize the importance of considering time-dependent material behavior and rock mass properties in design, especially near tunnel–gallery intersections.

Acknowledgements



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Three-dimensional finite element simulations

- Elastic lining higher pressures (see all Figures);
- Long-term higher pressures (see all Figures);
- In general, the lining pressures remain close to the reference value $p(\theta = 90^\circ)$, except in the range $0^\circ < \theta < 60^\circ$, where a significant increase is observed (see Fig. (b));
- Horizontal ovalization tunnel wall induced higher pressures at the lateral sides of tunnel wall ($\theta = 0^\circ$ and 180°);
- The friction angle significantly affect pressure distribution in S2, particularly for $30^\circ < \theta < 90^\circ$ at end of construction;
- Higher friction angles decrease pressures above the gallery entrance. More differences at the end of construction;

Research problem

- The excavation of a transverse gallery significantly modifies the stress field around the main tunnels;
- This modification causes an overload on the main tunnel linings;
- Quantifying this additional load is essential for defining structural reinforcements in this region and ensuring the safety of the tunnel system.;
- Therefore: How does the construction of the gallery affect the distribution of pressures on the lining?

Preliminary verification

Chortis and Kavvadas (2023):

- Generalized Hoek–Brown criterion
- Shotcrete lining: shell elements

Present study:

- Hoek-Brown criterion approximated by the Mohr-Coulomb criterion:

$$\text{Cohesion } (c) = 0.14 \text{ MPa}$$

$$\text{Friction angle } (\phi) = 28.72^\circ$$

- Shotcrete lining: solid elements

Geometry:

- $R_t/R_g = 1.33$
- $d_1 = 8R_t$

Rockmass model data:

- $E_m = 562 \text{ MPa}$
- $\nu_m = 0,3$
- $\sigma_v = \sigma_h = 2 \text{ MPa}$

Lining model data:

- $E_{c_{28}} = 20 \text{ GPa}$
- $\nu_c = 0,2$

Preliminary verification

The purpose is to verify the model's ability to simulate the **internal forces** induced in the tunnel lining by the transverse gallery.

$$N_C^* = \frac{N_C}{\sigma_v D}$$

S/D is the normalized longitudinal distance from the center of the intersection zone

